DATA MINING:

Data mining is the process of **extracting useful patterns, knowledge, or insights from large sets of data**. It helps organizations make decisions, predict trends, and discover hidden relationships in data

Data mining is the process of analyzing large sets of data to **discover hidden patterns, relationships, and useful knowledge** that can help in decision-making, predictions, and problem-solving. It involves collecting, cleaning, integrating, and transforming data, followed by applying techniques like **classification, clustering, regression, association rule mining, and anomaly detection** to extract meaningful insights. Data mining is widely used in **business, healthcare, finance, e-commerce, and education** to improve efficiency, forecast trends, detect fraud, and understand customer behavior.

Key points about data mining:

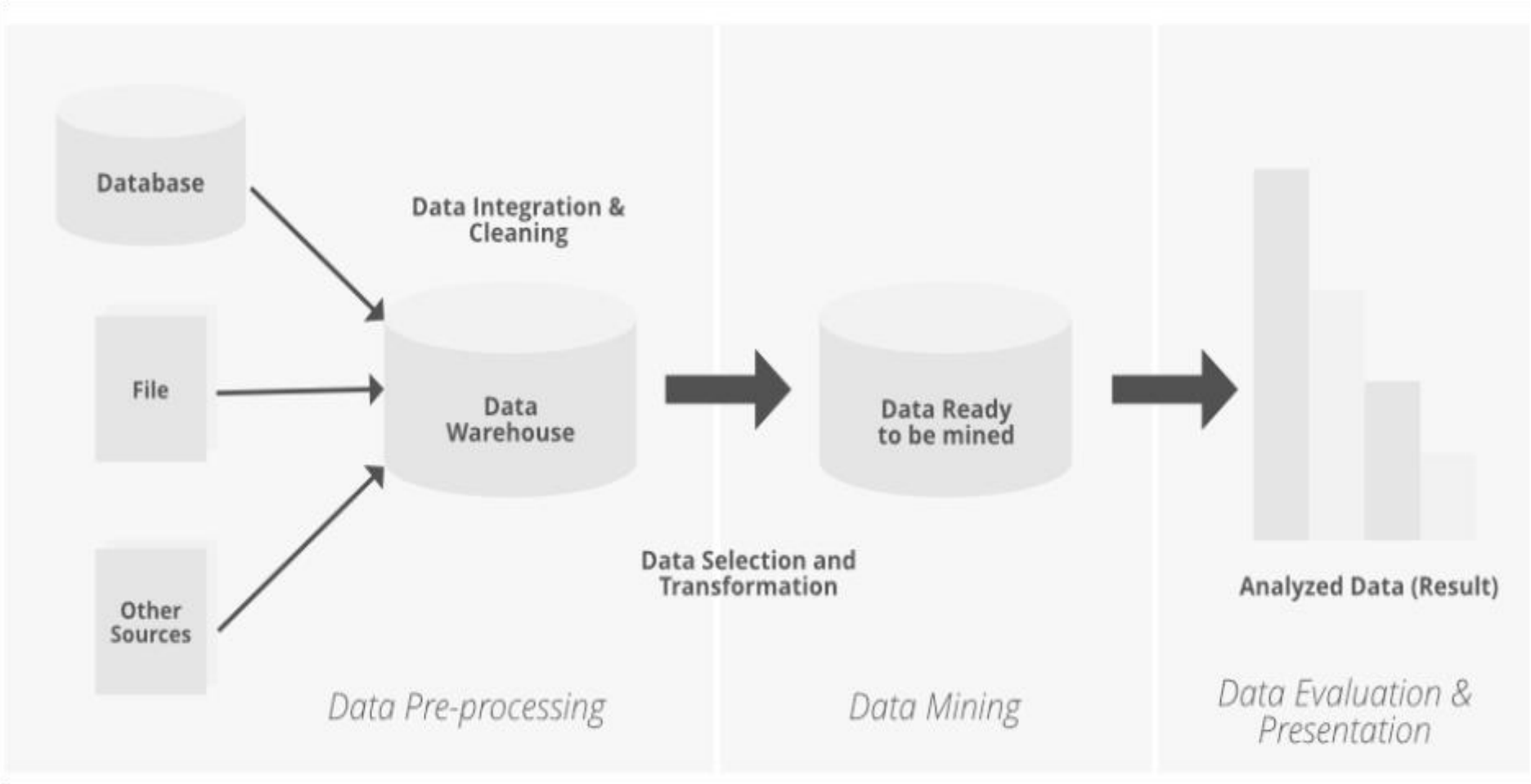
- It works on large datasets from databases, data warehouses, or other sources.
- It uses techniques from **statistics, machine learning, and database systems**.
- It can find patterns such as associations, trends, anomalies, or clusters.
- It helps in **decision-making, forecasting, and improving business processes**.

PURPOSE:

Purpose of Data Mining

The main purpose of data mining is:

- **Knowledge Discovery:** Find hidden patterns and insights in data.
- **Decision Support:** Help businesses and organizations make informed decisions.
- **Prediction:** Forecast future trends, like sales or customer behavior.
- **Classification:** Categorize data into meaningful groups.
- **Anomaly Detection:** Identify unusual data points or outliers.


Database
File
Other Sources
Data Integration & Cleaning
Data Warehouse
Data Selection and Transformation
Data Ready to be mined
Analyzed Data (Result)
Data Pre-processing
Data Mining
Data Evaluation & Presentation

Steps in Knowledge Discovery from Data (KDD)

Knowledge discovery from data (KDD) is a multi-step process for extracting useful insights. The following are the key steps involved:

- **Data Selection:** Identify and select relevant data from various sources for analysis.
- **Data Preprocessing:** Clean and transform the data to address errors and inconsistencies, making it suitable for analysis.
- **Data Transformation:** Convert the cleaned data into a form that is suitable for data mining algorithms.
- **Data Mining:** Apply data mining techniques to identify patterns and relationships in the data, selecting appropriate algorithms and models.
- **Pattern Evaluation:** Evaluate the identified patterns to determine their usefulness in making predictions or decisions.
- **Knowledge Representation:** Present the patterns in a way that is understandable and useful for decision-making.
- **Knowledge Refinement:** Refine the knowledge obtained to improve accuracy and usefulness based on feedback.
- **Knowledge Dissemination:** Share the results in an easily understandable format to aid decision-making.

- **Clustering:** Clustering is a data mining technique that groups similar data points together into clusters so that items in the same cluster are more alike, and items in different clusters are less similar. It is an **unsupervised method**, meaning it does not need predefined categories. Clustering is commonly used in **customer segmentation, market analysis, and image processing**.

- Clustering is a **data mining technique** that groups similar data points together.
- It is an **unsupervised method**; no predefined categories are needed.
- Items in the **same cluster are similar**, while items in different clusters are dissimilar.
- Common algorithms include **K-Means, Hierarchical Clustering, and DBSCAN**.

Association analysis looks for patterns where certain items or conditions tend to appear together in a dataset. Association rule mining is a data mining technique used to **discover relationships, patterns, or connections between items in large datasets**, especially in market basket analysis. It helps identify which items are frequently bought or used together.

- It is a **data mining technique** used to find relationships between items in large datasets.
- It works on **IF–THEN rules**, such as “If a customer buys bread, then they also buy butter.”
- Helps in **market basket analysis**, product recommendations, and cross-selling.

Classification – Definition

- Classification is a data mining technique that **assigns data into predefined categories or classes** based on their characteristics. It uses historical labeled data to build a model and then predicts the class of new, unseen data.

•Classification is a **data mining technique** used to place data into predefined classes or categories.
•It uses **labeled training data** to learn patterns (supervised learning).

•It helps in **decision-making**, pattern recognition, and future predictions.
•Widely used in **banking, marketing, healthcare, and email filtering**.

- ★ **Regression – Definition**
- Regression is a data mining and statistical technique used to **predict continuous or numerical values** by finding the relationship between one dependent variable and one or more independent variables. It helps in forecasting and estimating future values based on past data.
- Regression is used to **predict future numerical values** like prices, sales, demand, or trends based on past data.

- ★ **1. Artificial Neural Network (ANN) – Definition**
- An Artificial Neural Network (ANN) is a **machine learning model inspired by the human brain**, made of interconnected artificial neurons that learn patterns from data. It can recognize complex relationships by adjusting the strength of connections during training.
- **Where ANN is used**
- **Image and face recognition**
- **Speech and voice recognition**
- **Medical diagnosis**
- **Fraud detection**
- **Stock market prediction**
- **Natural Language Processing (NLP)**

- ★ **2. Outlier Detection – Definition**
- Outlier detection is a technique used to **identify data points that are very different or unusual** compared to the rest of the dataset. These points, called outliers, may indicate errors, fraud, or rare events.
- **Where Outlier Detection is used**
- **Fraud detection** in banking
- **Network intrusion detection** (cybersecurity)
- **Detecting faulty sensors** in IoT systems
- **Quality control** in manufacturing
- **Identifying unusual medical test results**

- **3. Genetic Algorithm – Definition**
- A Genetic Algorithm is an optimization technique inspired by **natural selection and evolution**. It solves problems by creating many possible solutions, selecting the best ones, and improving them over several generations (survival of the fittest).
- **Where Genetic Algorithms are used**
- **Scheduling problems** (like exam timetables or job scheduling)
- **Optimization in engineering**
- **Robot path planning**
- **Game strategy development**
- **Traveling Salesman Problem (TSP)**
- **Machine learning feature selection**